

STAT 412 HW07 — Question 12

Problem

It is claimed that automobiles are driven on average more than 18,000 kilometers per year. To test this claim, 110 randomly selected automobile owners are asked to keep a record of the kilometers they travel. Would you agree with this claim if the random sample showed an average of 18,330 kilometers and a standard deviation of 3700 kilometers? Use a P-value in your conclusion.

The conclusion portion asks whether to reject H_0 , whether there is or is not sufficient evidence to agree with the claim, and whether H_0 would be rejected for significance levels greater or less than the P-value.

Solution

The claim is “more than 18,000,” so the alternative is right-tailed:

$$H_0 : \mu = 18000, \quad H_a : \mu > 18000.$$

This corresponds to choice D.

The population standard deviation is not given, so use a one-sample t statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{18330 - 18000}{3700/\sqrt{110}}.$$

Compute the denominator:

$$\frac{3700}{\sqrt{110}} \approx 352.78.$$

Then

$$t = \frac{330}{352.78} \approx 0.93542.$$

Rounded to two decimals, the test statistic is

$$t = 0.94.$$

With $df = 109$, the right-tail P-value is about

$$P(T_{109} > 0.93542) \approx 0.1758.$$

Using the t table, this is approximately in the range

$$0.15 < P\text{-value} < 0.20.$$

Since the P-value is large, do not reject H_0 . The sample mean is above 18,000, but not by enough to statistically support the “more than 18,000” claim. In general, reject H_0 for significance levels greater than the P-value.

Final Answer

Choice D: $H_0 : \mu = 18000$; $H_a : \mu > 18000$. Test statistic $t = 0.94$; P-value ≈ 0.176 (table range $0.15 < P < 0.20$). Do not reject H_0 ; there is not sufficient evidence to agree with the claim. Reject for significance levels greater than the P-value.